Evaluating the Impact of Congestion Pricing on Taxi Demand and Air Quality in Manhattan's CBD

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Github repo with commit

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1 Introduction

New York, specifically the Manhattan CBD, is an area accustomed to high volumes of traffic. To combat this, the Metropolitan Transportation Authority (MTA) introduced a congestion fee for most vehicles going in and out of the area, inclusive of Yellow Taxis, and High Volume For-Hire Vehicles (HVFHV). Despite having only been implemented on January 5, 2025, the federal government has already threatened to terminate this initiative. Critics have also argued that congestion pricing is a cynical cash grab by mismanaged MTA, and have claimed this toll to be elitist (The Guardian, 2025). This raises the question of whether the policy is worth sustaining. Specifically, what was the impact of the January 2025 CBD congestion fee on for-hire and taxi traffic volumes, and did it correspond to measurable changes in air pollution levels in Manhattan's CBD?

In this paper, we will implement machine learning models to predict the CBD trip counts, and the air quality using two methods; linear regression and random forest regression. These measures are of note, as some of the main motivations for introducing this congestion fee was to reduce traffic and improve air quality (The Guardian, 2025). This paper aims to develop a comprehensive analysis to guide the federal government and other stakeholders related to policy making into deciding whether to axe the initiative or not.

1.1 Datasets

A variety of datasets were used for our investigation. The main dataset used was the TLC Taxi Trip Record data and the TLC HVFHV Trip Record data from the NYC Taxi and Limousine Commission, which included trip records (datetime of pickup/drop off, fare amount, etc.) from Yellow Taxis and HVFHVs respectively. Daily estimated subway ridership was also a resource, which noted the amount of people that tapped onto the NYC subway each day. Quality data was used, from the NYC environment and health data portal. This dataset outlines the real-time Air Quality: PM2.5 in NYC, containing hourly measurements of PM2.5, in micrograms per cubic meter of air (µg/m3) (NYC Department of Health and Mental Hygiene, 2024).

In these datasets, only records within the 6 month interval from November 2024 to April 2025 were used. Since the congestion pricing began in January 5, 2025, this interval captures before and after it was implemented. The following table provides a summary of all the datasets:

Datasets	Instances	No. features
TLC Yellow Taxi Trip Record Data	22,483,319	19
TLC HVFHV Trip Record Data	121,092,373	24
Real-Time Air Quality: PM2.5 in NYC	55,272	4
MTA Subway Ridership	1,887	15

Table 1: Datasets shape

2 Preprocessing

Given that these datasets are large and varied, it was imperative that the data was preprocessed to ensure consistency before the statistical models were implemented.

2.1 Yellow Taxi and High Volume for Hire Vehicle Data

- Column names across the Yellow Taxi and HVFHV datasets were standardized to prepare them later to be joined.
- Timestamps, distances, fares, tips, tolls, and fees were converted into a consistent format.
- Basic quality rules were applied; this involved ensuring numeric values, such as passenger counts and fare amounts were non-negative. Extremely high/low values were also omitted (i.e. for distance/durations).
- Trips that were not dispatched within the Manhattan CBD area were filtered out. This is defined as trips with either a pick up or drop off location in the area defined as the Manhattan CBD.
- Between the datasets, it was important to denote the location IDs of interest to focus the investigation. The Manhattan CBD area is defined as insert citation. In our investigation, we denoted these areas as: 50, 48, 163, 230, 161, 162, 229, 233, 170, 164, 100, 186, 68, 90, 234, 107, 137, 224, 246, 158, 249, 113, 114, 125, 211, 144, 148, 232, 45, 231, 13, 261, 209, 87, 88, 12, 4, 79.

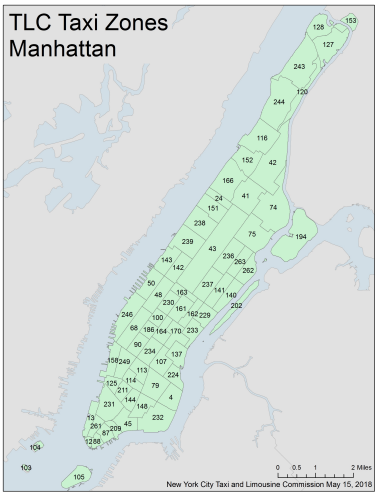


Figure 1: TLC Taxi Zones in Manhattan. Source: New York City Taxi and Limousine Commission.

2.2 Air Quality Data

- The hourly file has a timestamp column that sometimes contains zero-width characters, RTL marks, or BOM markers that break parsing.
- For the air quality data, we restricted our analysis to monitors located within the Manhattan CBD, as identified from the official NYC Environment and Health Data Portal.
- The hourly data is then aggregated to daily means per site, tags by month
- Results are combined into one DataFrame using unionByName
- The distribution of the datasets were plotted to detail the spread, ensuring no obvious outliers.

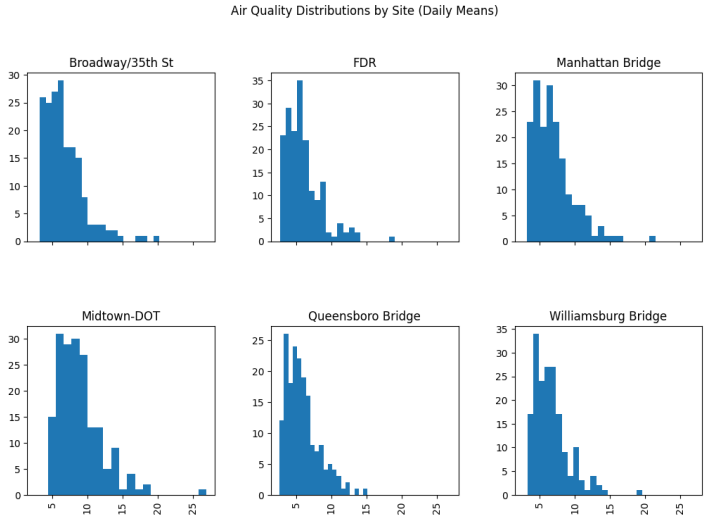


Figure 2: Distribution of daily mean $PM_{2.5}$ levels across CBD monitoring sites.

2.3 Daily Estimated Subway Ridership

- The dates and datetimes were made consistent.
- The datasets were checked for missing values to ensure completeness.
- Since this dataset was essentially two datasets combined, (one hourly and one daily), the hourly data was aggregated to daily values in order to reduce noise and ensure a consistent level of granularity with the other datasets.
- The hourly data only had 3 features, as opposed to the 15 in the one recorded daily. Thus, the column names had to be normalized, and features that were not present in either were omitted, as it did not seem relevant to our investigation.
- Once the data had been aggregated by daily average, a histogram was plotted to visualize the spread of the data A statistical method of using $Q1, Q3 \pm 1.5 \times IQR$ was utilized to further reiterate the absence of any outliers.

2.4 Feature Engineering and Selection

Once the data had been cleaned, features were created, such as fare per mile, fare per minute, a weekday/end flag, and post-policy indicator (whether the trip happened after the congestion pricing

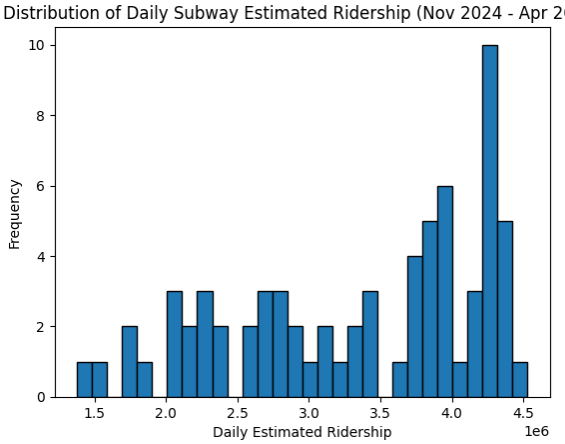
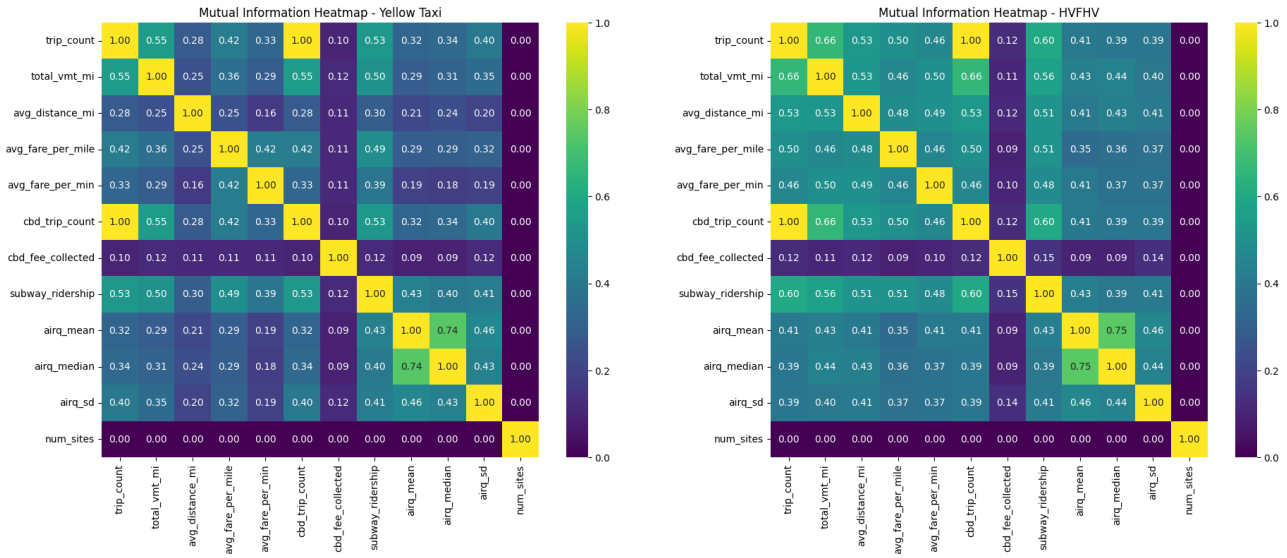


Figure 3: Distribution of daily estimated subway ridership (Nov 2024–Apr 2025).

policy date). Trips that start or end in the CBD were also flagged.

Heatmaps were then computed, to visualize the degree of correlation. Normalized mutual information was used due to its ability to capture non-linear relationships between variables, which is useful for feature selection in the implementation of the random forest regression (Rossi & Ahmed, 2018). It is important to normalize the mutual information values as it places these values within a standardized range.



(a) Yellow Taxi: Pearson/MI summary

(b) HVFHV: Pearson/MI summary

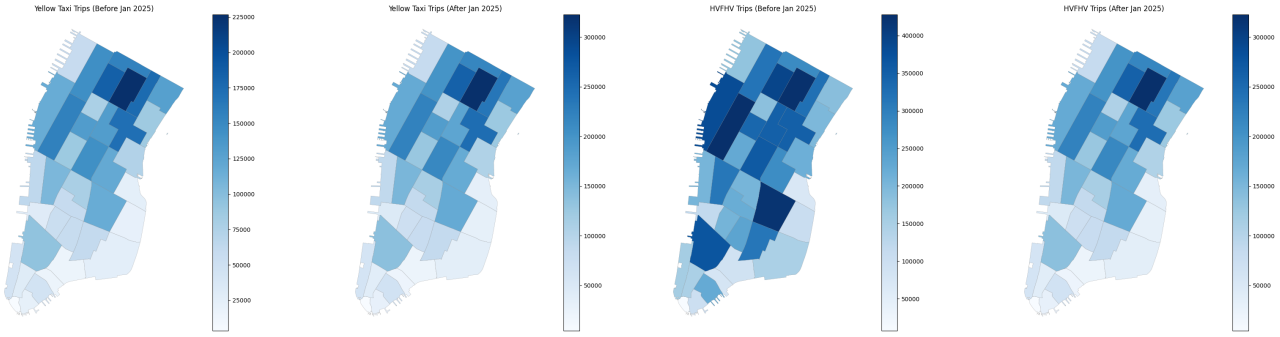
Figure 4: Feature-screening heatmaps used for model design.

For the **air quality prediction**, the following variables were used as predictors: `total_vmt_mi`, `trip_count`, `avg_fare_per_mile`, `cbd_trip_count`, and `subway_ridership`.

For the **trip count prediction**, the predictors were: `total_vmt_mi`, `avg_fare_per_mile`, `avg_fare_per_min`, `subway_ridership`, and `airq_mean`.

3 Analysis and Geospatial Visualisation

We can see yellow taxis were not as affected by congestion pricing as the number of trips dispatched before and after are relatively similar. This is supported by external reports: "Concerns that riders would abandon taxis to avoid the added $0.75 congestion fee have also proven unfounded... (The Guardian, 2025)."

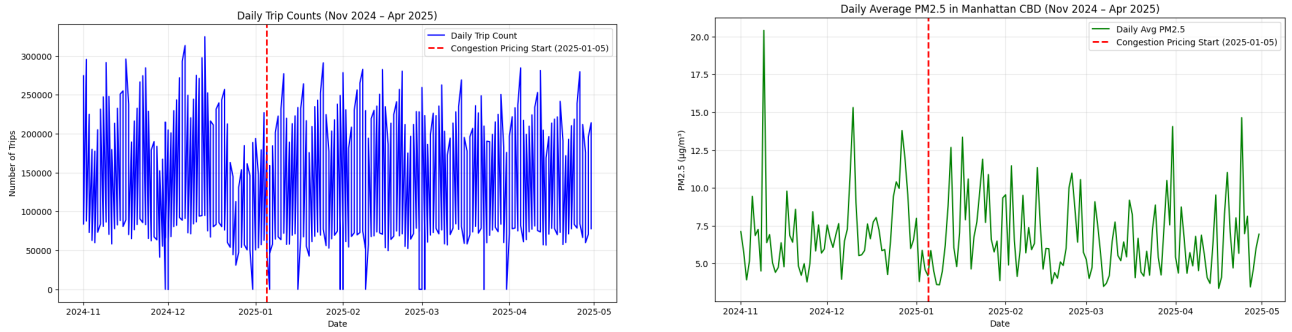


(a) Yellow Taxi Trips (Before vs After Jan 2025)

(b) HVFHV Trips (Before vs After Jan 2025)

Figure 5: Comparison of Yellow Taxi and HVFHV trip distributions before and after congestion pricing.

HVFHV dispatches within the CBD changed drastically before and after congestion pricing. Fig 5(b) suggests a significant decrease in HVFHVs in the Manhattan CBD after the toll was implemented. In fact, "Since January, an average of 70,000 fewer vehicles have entered the congestion zone each day, according to the MTA (The Guardian, 2025)."



(a) CBD Trip Trends (Before vs After Jan 2025)

(b) Air Quality Trends (Before vs After Jan 2025)

Figure 6: Trends in CBD trip counts and air quality before and after congestion pricing.

As shown in Figure 6a, after congestion pricing was implemented, the peaks of daily trip counts were generally lower than before, suggesting a possible correlation between the policy and reduced traffic in the CBD. Figure 6b shows a slight downward trend in $PM_{2.5}$, though levels rise again in April. While the first spike might appear to be an outlier, formal outlier detection ($Q1, Q3 \pm 1.5 \times IQR$) did not identify any extreme values.

4 Modelling

Linear Regression and Random Forest Regression. points to dicsuss: model implementation.

4.1 Linear Regression

The modelling pipeline for Linear Regression was implemented in PySpark. Key steps were:

- **Data cleaning and parsing:** The raw panel data was first cleaned, with dates parsed across multiple formats. Invalid dates were flagged and inspected. Services outside `yellow` and `hvfhv` were removed.
- **Policy flags:** Difference-in-Differences (DiD) terms were added: a treatment indicator (`is_yellow`), a post-policy indicator (`post_policy`), and their interaction (`policy_x_yellow`).
- **Feature preparation:** All numeric variables not serving as identifiers or labels were retained as candidate features. Essential variables (e.g., outcome, service, date) were required to be non-null, while missing values in the other features were imputed with the median. Features were then standardised to mean zero and unit variance.
- **Cross-validation:** To respect temporal ordering, an expanding-window time series cross-validation was implemented. For each fold, earlier data were used to train and the immediately following period to validate. A grid of LASSO penalty values (α) was tested.
- **Model fitting:** The best penalty was chosen based on lowest average RMSE across folds. A final LASSO model (Linear Regression with ℓ_1 regularisation) was trained on the full dataset using this optimal α. Coefficients were inspected to assess the relative contribution of each standardised feature.
- **Diagnostics:** Performance was reported via R^2 and RMSE, both in-sample and across cross-validation folds.

4.2 Random Forest Regression

The Random Forest model was implemented in scikit-learn on the Pandas version of the panel data. It was initially implemented using PySpark, but excessive run time necessitated the use of Pandas for this section. The main motivations were to capture potential non-linear effects that linear models may not fully represent.

- **Data preparation:** The Spark panel was converted to a Pandas DataFrame, dates were parsed, and only `yellow` and `hvfhv` services were retained. The same DiD flags as above were constructed. Rows were sorted chronologically to preserve temporal order.
- **Feature set:** Predictor variables included the policy indicators, transport demand metrics (e.g., `trip_count`, `total_vmt_mi`, `cbd_trip_count`), price-related variables (e.g., `avg_fare_per_mile`, `avg_fare_per_min`), subway ridership, and air quality. These were selected based on their mutual information scores with the outcome.
- **Baseline model:** A baseline Random Forest was fit with sensible defaults (400 trees, $\sqrt{\text{features}}$ per split, minimum leaf size of 2). Train/test splitting was chronological (80% train, 20% test).
- **Hyperparameter tuning:** GridSearchCV with a time series split (five folds) was used to tune the number of trees, maximum depth, minimum split/leaf sizes, and feature subsampling rate. The grid was scored by RMSE, and the best configuration was selected and refit on the full training span.
- **Evaluation:** Both baseline and tuned models were evaluated on the holdout set, reporting R^2 and RMSE. Feature importances were extracted to interpret the most influential predictors.

5 Discussion

In this section, we will evaluate the models and discuss their implications.

Model	Target	R^2	RMSE	Key Features (Top Coeffs)
LASSO Regression	Trip Count	0.990	8,458	+ total_vmt_mi (95,401) + cbd_fee_collected (31,240) + policy_x_yellow (25,123) – post_policy (-30,000) – airq_mean (-17,916) – avg_distance_mi (-15,713)
LASSO Regression	Air Quality (PM2.5 mean)	≈ 0	2.62	all coefficients ≈ 0
Random Forest	Trip Count	0.979	10,095	total_vmt_mi (0.96) avg_fare_per_min (0.02) is_yellow (0.01)
Random Forest	Air Quality (PM2.5 mean)	-0.21	2.97	subway_ridership (0.25) cbd_trip_count (0.21) trip_count (0.20)

Table 2: Model evaluation metrics and most influential features. Positive coefficients (LASSO) and high feature importance values (RF) indicate stronger associations.

5.1 Linear Regression

The LASSO regression achieved strong predictive performance for `trip_count`, with an R^2 of 0.99. The most influential predictors were indicators of vehicle activity and pricing dynamics, such as `total_vmt_mi`, `cbd_fee_collected`, and `avg_fare_per_mile`. Policy-related terms (`post_policy` and `policy_x_yellow`, which is the interaction term) also featured prominently, suggesting that congestion pricing itself contributed measurably to changes in trip volumes.

When applied to `airq_mean`, however, the model performed poorly ($R^2 \approx 0$). All coefficients shrank toward zero under regularization, indicating no consistent linear relationship between vehicle trip metrics and daily $PM_{2.5}$ levels. This suggests that air quality is influenced more heavily by external factors such as weather, cross-borough traffic, industrial emissions, etc., rather than by taxi and HVFHV activity alone.

5.2 Random Forest Regression

The Random Forest model confirmed the strong predictive signal for `trip_count`, achieving an R^2 of 0.98 on the set with tuned hyperparameters. On the contrary, the random forest regression did not perform as well in predicting the air quality means. The model yielded an R^2 of -0.21 on th best hyperparameters.

5.3 Interpreting RMSE Values

In addition to R^2, the root mean squared error (RMSE) provides insight into the scale of typical prediction errors. For `trip_count`, the LASSO regression achieved an RMSE of 8,458, while the Random Forest returned 10,095. Given that daily trip volumes in Manhattan's CBD number in the hundreds of thousands, these errors represent a relatively small proportion of overall demand. This

indicates that both models deliver predictions that are accurate enough to capture the overall traffic patterns, with LASSO slightly outperforming Random Forest.

For `airq_mean`, the situation is different. The RMSE values of 2.62 (LASSO) and 2.97 (Random Forest) should be compared against typical daily $PM_{2.5}$ levels in Manhattan, which often range between 5 and 15 $\mu g/m^3$. In this context, the errors correspond to roughly 20–40% of the daily average. This reinforces the idea that neither model was able to capture meaningful predictive structure in air quality outcomes, as the variability is dominated by external environmental factors rather than taxi/HVFHV activity.

6 Recommendations

The evidence suggests that congestion pricing in Manhattan's CBD has achieved a measurable reduction in traffic volumes, particularly among HVFHV dispatches, while leaving yellow taxi activity largely unaffected. This aligns with the policy's goal of reducing overall congestion without disproportionately harming taxi drivers. However, our modelling results indicate that the effect on air quality is limited: neither linear nor non-linear models could reliably predict $PM_{2.5}$ levels from trip counts alone, suggesting that external factors such as weather patterns and emissions from other sources have a greater influence.

Based on these findings, we recommend that:

- **Continue the congestion pricing program:** The policy is clearly effective in reducing HVFHV traffic, which is a key driver of congestion in the CBD. Stable yellow taxi activity further suggests that the measure does not undermine this sector.
- **Avoid premature federal intervention:** While air quality improvements are less conclusive, congestion reduction itself is a valuable policy outcome with downstream benefits for mobility, safety, and potentially longer-term emissions.
- **Integrate supplementary measures:** To achieve the air quality objectives more directly, congestion pricing should be paired with complementary policies such as expanding electric vehicle adoption, or subsidized public transport use within the zone.

The federal government should not terminate congestion pricing prematurely. Instead, the program should be sustained and refined, as it is already delivering on its central aim of reducing CBD congestion while maintaining stable taxi livelihoods.

7 Conclusion

This study evaluated the impact of Manhattan's January 2025 congestion pricing initiative on taxi and HVFHV traffic volumes and air quality in the CBD. Using both LASSO regression and Random Forest regression, we found consistent evidence that the policy reduced trip counts, especially among HVFHVs. However, neither model identified a strong relationship between trip volumes and $PM_{2.5}$ levels, which emphasizes the complexity of urban air quality and the influence of external environmental factors.

The findings suggest that congestion pricing is achieving its primary goal of traffic reduction without negatively affecting yellow taxi drivers. Although its effect on air quality remains uncertain, this should not be interpreted as policy failure. The initiative is worth sustaining, with future research encouraged to incorporate additional variables such as meteorological conditions and cross-borough traffic patterns to get a better gauge on long-term environmental outcomes.

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